MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG3DSCECT200 Analog Electronics

Duration: 1.5 Hours **Maximum Marks: 50**

**Remember (K), Understand (U), Apply (A), Analyze (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions
Answer **All** Questions
Each question carries 1 mark

1	How many stages are there in a BJT? a)1 b) 2c) 3 d) 4	U	1
2	In a BJT which region is thin and lightly doped? a) Base b) Emitterc)Collector d) none of the above	U	1
3	The quiescent point in a BJT is also referred to as a) Operating point b) Cut off point c) Saturation points d) Maximum point	U	1
4	Voltage divider circuit is based on a)Faradays law b) Ohms Lawc) Kirchoff's law d) None of the above	U	1
5	An example of Linear IC is a) Microprocessor b) IC 741 c) flip flop IC d) Logic Gate IC	U	2
6	A typical op amp has how many internal stages? a) 1 b) 2c)3 d)4	U	2
7	In a typical op amp IC 741 what is the function of pin No 6? a) Input b) Offset nullc) output d) Vcc	U	2
8	Unit of CMRR is a) Volts b) Amperec) Hz d) dB	U	2
9	What is the primary function of 555 IC? a) Amplification b) Rectificationc) Regulation d) Timing and Oscillation	K	3
10	What is the primary function of IC 566? a) To amplify signals b) To generate voltage-controlled oscillations c) To regulate voltage c) To generate square wave	K	3

1 * 10 = 10

Part B

Short Answer Questions
Answer any **4** Questions
Each question carries 5 marks

11	Explain the term operating point in BJT design.	U	1
12	Define the two Barkhausen criteria for oscillations.	U	1
13	Draw the pin diagram of the general-purpose Op amp IC 741	U	2

14	Explain the slew rate characteristics of an Op amp.	U	2
15	Draw the circuit of an Op amp integrator and draw its input and output waveforms.	An	3
16	Draw the circuit of a zero-crossing detector using an Op amp and draw its input and output waveforms.	An	3

4 * 5 = 20

Part C

Essay Questions

Answer any **2** Questions

Each question carries 10 marks

17	Explain the three modes of operation of a BJT?	U	1
18	Explain the criteria behind the operation of a sinusoidal oscillator?	U	1
19	Explain the block diagram representation of an Op amp?	U	2
32	Draw and explain the operation of an RC phase shift oscillator using Op amp?	An	3

2 * 10 = 20